Weekly Progress Report

Amity School of Engineering & Technology

B.Tech: Program CSE
WPR No 1
Duration: 4/05/26 to 10/05/26

Students' Weekly Progress Report (WPR) for NTCC May-June 2026

To be filled by Students	
Students Name	1. Manav Kasnia 2. Rishank Veermanya
Roll no.	1. 357 2. 356
Enrollment no.	1. A2305224357 2. A2305224356
Project Title finalized, if Yes, give name, if NO, give reason	Prithvi Pulse
Synopsis submitted	Yes
Literature review	This a ML based model that can give a city a score(CSI) based on some factors as Pollution, Population Density, Climate Change, Traffic and more. So that we can work on the things that are not well to maintain a good life style. This will gradually increase the life span of human. But we have to take necessary steps.
Technical & Economical Feasibility	**Data** — All the data we need (AQI, traffic, weather, population) already exists online. Free APIs cover all Indian cities. No data problems. **Tech stack** — React for frontend, Python for backend, scikit-learn for ML. All free, all beginner-friendly, all well-documented. Nothing exotic. **ML part** — We're not doing anything crazy. A weighted formula + Random Forest + K-Means. Runs on your laptop. No GPU needed. **Timeline** — 60 days for this scope is comfortable. Not rushed. **Risk** — Low. The only real risk is free API rate limits, which our usage won't hit. Economy — Almost free to build
Bill of Material	0
Project Progress Schedule (PERT Chart)	**Task** / **What it is** / **Days** / **Depends on** T1 / Problem statement + scope / 1–5 / Nothing (start here) T2 / Literature review / 3–10 / T1 T3 / CSI formula design / 6–14 / T1, T2 T4 / API registration + testing / 4–14 / T1

11/5/26

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<tr><td></td><td>
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<tr><td>T5</td><td>Backend + data pipeline</td><td>15–28</td><td>T3, T4</td></tr>
<tr><td>T6</td><td>Frontend dashboard</td><td>29–42</td><td>T5</td></tr>
<tr><td>T7</td><td>Historical data collection</td><td>28–42</td><td>T5</td></tr>
<tr><td>T8</td><td>ML validation</td><td>43–49</td><td>T7</td></tr>
<tr><td>T9</td><td>Paper writing</td><td>42–56</td><td>T8</td></tr>
<tr><td>T10</td><td>Deploy app + submit paper</td><td>57–60</td><td>T6, T9</td></tr>
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<tr><td>Design of critical components</td><td>CSI formula and API design is important yet to do</td></tr>
<tr><td>Fabrication work (give %)</td><td>40%</td></tr>
<tr><td>Experimental work (give %)</td><td>35%</td></tr>
<tr><td>Result and Analysis</td><td>CSI Scores for 6 Indian cities Delhi will likely score highest (extreme AQI + density). Mumbai high on traffic. Bangalore high on traffic + weather. Chennai moderate. Pune and Hyderabad comparatively lower.
Feature importance finding Random Forest will tell us whether AQI truly deserves 30% weight or if traffic should be higher. This delta between our assumed weights and ML-computed weights is the key research finding.
City stress archetypes (K-Means) Cities will cluster into types — pollution-dominant cities (Delhi), traffic-dominant cities (Bangalore, Mumbai), balanced stress cities (Pune, Hyderabad). This is a novel contribution.
Benchmark correlation If CSI scores correlate well (r > 0.7) with Mercer or EIU livability rankings for Indian cities, it validates the entire approach.
Temporal patterns CSI will spike on weekday mornings (traffic), winter months (AQI in Delhi), and monsoon season (weather stress). These patterns become discussion points in the paper.</td></tr>
<tr><td>Report writing</td><td>25%</td></tr>
<tr><td>Signature of students</td><td>Rishank</td></tr>
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Work done in this week

Task 1 problem statement and research on some data sets API is done.

To be filled by Guide (strike off whichever is not applicable)
Performance of students is satisfactory
Performance of students is unsatisfactory
A warning to be issued to student(s) (Name)
Student was not well (Name)
Date Signature of Guide 11/5/26

Date: 10 May 2026.